

# Dataset Details

|                                |                                                                                                                                                                                                                                                           |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                    | Diabetes 130-US Hospitals for Years 1999-2008                                                                                                                                                                                                             |
| <b>Source</b>                  | UCI Machine Learning Repository (donated 2014-05-02)                                                                                                                                                                                                      |
| <b>Link</b>                    | <a href="https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008">https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008</a>                                                                 |
| <b>DOI</b>                     | 10.24432/C5230J                                                                                                                                                                                                                                           |
| <b>Licence</b>                 | Creative Commons Attribution 4.0 International (CC BY 4.0), as stated on the UCI page                                                                                                                                                                     |
| <b>Citation</b>                | Strack, B., DeShazo, J., Gennings, C., Olmo, J., Ventura, S., Cios, K., & Clore, J. (2014). Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records. <i>BioMed Research International</i> , 2014. |
| <b>Format</b>                  | Two CSV files: <code>diabetic_data.csv</code> (101,766 rows, 50 columns) and <code>IDs_mapping.csv</code> (three lookup tables stacked in one file)                                                                                                       |
| <b>Purpose in this project</b> | Teaching data for an end-to-end healthcare data system: ETL, quality control, a relational database, a secured REST API, KPI analytics and a dashboard. <b>Educational use only.</b>                                                                      |

## 1. What one row means

One row is one hospital **encounter** (an inpatient stay) of a diabetic patient at one of 130 US hospitals, 1999-2008. `encounter_id` is unique (101,766 distinct values). `patient_nbr` repeats: there are **71,518 distinct patients**, 16,773 of whom appear more than once (the most frequent has 40 encounters). Patients and encounters are therefore never used interchangeably in this project.

## 2. Key fields

| Field                                                                                                                                                                                                                         | Meaning                               | How it is used                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <code>encounter_id</code> , <code>patient_nbr</code>                                                                                                                                                                          | Source keys (BIGINT, never generated) | Primary keys of <code>encounters</code> and <code>patients</code>                                                |
| <code>age</code>                                                                                                                                                                                                              | Ten brackets [0-10) ... [90-100)      | Stored as <code>age_group</code> plus <code>age_order</code> 1-10; always sorted by <code>age_order</code>       |
| <code>race</code> , <code>gender</code>                                                                                                                                                                                       | Demographics (sensitive)              | Stored only on <code>patients</code> ; hidden from the analyst role; never a dashboard breakdown, never exported |
| <code>admission_type_id</code> , <code>admission_source_id</code> , <code>discharge_disposition_id</code>                                                                                                                     | Coded categories                      | Foreign keys to lookup tables loaded from <code>IDs_mapping.csv</code> (8, 25 and 30 rows)                       |
| <code>medical_specialty</code> , <code>payer_code</code>                                                                                                                                                                      | Admitting specialty, payer            | Lookup tables; NULL when missing                                                                                 |
| <code>time_in_hospital</code>                                                                                                                                                                                                 | Length of stay in days (1-14)         | Length-of-stay KPIs                                                                                              |
| <code>num_lab_procedures</code> , <code>num_procedures</code> , <code>num_medications</code> , <code>number_outpatient</code> , <code>number_emergency</code> , <code>number_inpatient</code> , <code>number_diagnoses</code> | Counts of services and prior visits   | Stored on <code>encounters</code> ; prior inpatient visits drive the utilisation analysis                        |

| Field                                                  | Meaning                                                | How it is used                                                   |
|--------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------------------|
| diag_1, diag_2, diag_3                                 | ICD-9 diagnosis codes (strings, include V and E codes) | Rows in encounter_diagnoses (position 1-3)                       |
| max_glu_serum, A1Cresult                               | Lab results; the text None means "test not performed"  | Stored as ENUMs; None is a real value                            |
| 23 drug columns (metformin ... metformin-pioglitazone) | No, Steady, Up, Down                                   | Long table encounter_medications holding only Steady / Up / Down |
| change, diabetesMed                                    | Medication change, diabetes medication prescribed      | med_changed, diabetes_med booleans                               |
| readmitted                                             | NO, >30, <30                                           | The outcome; see section 4                                       |

## 3. Preprocessing performed (every rule)

### 3.1 Reading

- ? is read as missing (keep\_default\_na=False, na\_values=['?', '']). The literal text None in max\_glu\_serum and A1Cresult is **kept**: with pandas defaults it would be turned into missing (96,420 and 84,748 rows), wrongly suggesting that most patients have no lab result.
- ICD-9 columns are read as text so that v and E codes and decimal points survive.

### 3.2 Columns removed

| Column               | Reason                           |
|----------------------|----------------------------------|
| weight               | 96.86% missing                   |
| examide, citoglipton | constant value (No) in every row |

### 3.3 Transformations

- Column names to snake\_case; change to med\_changed (Ch = true), diabetesMed to diabetes\_med (Yes = true).
- age to age\_group plus age\_order (1-10).
- diag\_1..3 to three rows per encounter at most, in encounter\_diagnoses.
- The 21 informative drug columns to long format; rows with No are **not stored** (120,054 prescribed cells instead of 2,137,086 rows, 94.4% of which would be "No").
- gender value Unknown/Invalid (3 rows) stored as Unknown and flagged (DQ12); never imputed.
- Race and gender are taken per patient: the first non-null value by lowest encounter\_id; a conflict inside a file is flagged (DQ13).

### 3.4 Data-quality rules (code: src/etl/quality.py; full table: docs/data\_contract.md section 8b)

| Rule | Check                                                                | Action                  |
|------|----------------------------------------------------------------------|-------------------------|
| DQ01 | key columns present, integer, positive                               | reject                  |
| DQ02 | duplicate encounter_id in the file                                   | keep first, reject rest |
| DQ03 | admission type, source or disposition id not in the reference tables | reject                  |

| Rule | Check                                                                                                        | Action                    |
|------|--------------------------------------------------------------------------------------------------------------|---------------------------|
| DQ04 | value outside the allowed set ( readmitted , age , gender , drug status, lab results, change , diabetesMed ) | reject                    |
| DQ05 | numeric ranges (stay 1-14; counts non-negative and below caps)                                               | reject                    |
| DQ06 | ICD-9 format (short numeric codes are valid: the source dropped leading zeros)                               | set code to NULL and flag |
| DQ07 | dates missing or not ISO YYYY-MM-DD                                                                          | reject                    |
| DQ08 | dates outside 1999-01-01..2008-12-31 (discharge up to 14 days later) or in the future                        | reject                    |
| DQ09 | discharge before admission                                                                                   | reject                    |
| DQ10 | date difference differs from time_in_hospital                                                                | reject                    |
| DQ11 | missing values per column                                                                                    | metric only               |
| DQ12 | gender Unknown/Invalid                                                                                       | keep as Unknown , flag    |
| DQ13 | conflicting race or gender for one patient in a file                                                         | keep first non-null, flag |

A file in which more than 20% of rows are rejected fails as a whole and loads nothing; otherwise good rows load and bad rows are written to `data/rejected/` with the rule and reason. On the genuine data every batch is rejected **0 rows**; the rules are demonstrated on a deliberately faulty batch (23 rows rejected, 5 corrected).

### 3.5 Derived fields

| Field                   | Definition                                                                           |
|-------------------------|--------------------------------------------------------------------------------------|
| readmitted_30d          | readmitted == '<30'                                                                  |
| any_readmission         | readmitted is <30 or >30                                                             |
| is_readmission_eligible | false when discharge_disposition_id is 11, 13, 14, 19, 20 or 21 (expired or hospice) |
| age_order               | position of the age bracket, 1-10                                                    |

## 4. The readmission rate

The 30-day readmission rate is `readmitted_30d` among eligible encounters divided by the number of **eligible** encounters. Patients who died or entered hospice could not be readmitted, so they leave the denominator (2,423 encounters). On the full data this is  $11,314 / 99,343 = \mathbf{0.113888}$  (11.39%). Dividing by all 101,766 encounters would give 0.111177, which understates the rate without raising any error. The rate is per encounter, not per patient.

## 5. SIMULATED fields (not in the source)

The source has **no dates**. To support trends, date-range queries and incremental loading, two fields are created:

| Field          | Rule                                                                                                                                                                                                                                                             |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| admission_date | <b>Simulated.</b> Seeded random dates ( <code>numpy.random.default_rng(seed=42)</code> ) between 1999-01-01 and 2008-12-31 with mild growth and winter seasonality, sorted and assigned in <code>encounter_id</code> order. Re-running produces identical files. |

| Field          | Rule                                                      |
|----------------|-----------------------------------------------------------|
| discharge_date | <b>Simulated.</b> admission_date + time_in_hospital days. |

Every dashboard, API response (`meta.dates_simulated = true`), report and document that shows a date says so. Monthly or yearly patterns are illustrative and must never be read as clinical or epidemiological findings.

## 6. Other limitations of the data

- **No hospital identifier:** hospital-level comparisons are impossible; admission type, admission source, discharge disposition and specialty are compared instead.
- **Missingness:** `medical_specialty` 49.08% and `payer_code` 39.56% missing; category comparisons describe only the recorded subset.
- **Repeated patients:** encounters of one patient are not independent.
- **Dated and observational:** 1999-2008 data; findings are associations, not causes, and do not describe current practice.
- **Coded unknown categories** (for example discharge disposition 18, admission source 17) are kept as unknown, not imputed.

## 7. Privacy note

The extract is de-identified research data released for public use. The system still treats it as sensitive: `race` and `gender` exist only on the `patients` table, are visible only to administrators and clinical operations staff, never appear in exports, analytics, logs or error messages, and analytics groups with fewer than 11 encounters are flagged `small_n`. Analysts see encounters without `patient_nbr`. Raw data files are not committed to the repository. The system is for education only and must not be used for clinical decisions.